

Algorithmic Control and Invisible Labor: How Food Delivery Platforms Shape Rider Poverty

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October 5, 2026

1 Introduction

The rapid growth of platform-mediated food delivery services, such as Meituan in China and UberEats in Australia, exemplifies the structural transformation of labor markets in the gig economy. These platforms emphasize flexibility and independence, portraying riders as entrepreneurs free to choose when and how much to work. Yet behind this rhetoric lies a regime of algorithmic management: a set of digital systems that allocate orders, monitor performance, and structure incentives. This essay argues that algorithmic management in food delivery not only undermines labor conditions by externalizing costs to riders and embedding information asymmetries, but also entrenches poverty by suppressing riders' net earnings and limiting their ability to achieve a basic standard of living. Using evidence from both China and Australia, the essay demonstrates that what appears as autonomy is in practice a new form of algorithmic control.

Existing research has increasingly examined the rise of algorithmic management in gig work. De Stefano (2016) argues that platform-mediated labor represents a "new employment relationship" in which control is exerted through data rather than direct supervision. Similarly, Vallas and Schor (2020) highlight how the so-called "sharing economy" reproduces classical forms of labor subordination behind the rhetoric of flexibility. Building on this literature, this essay focuses specifically on food delivery platforms in China and

Australia to show how algorithmic management translates digital control into material poverty among riders.

2 Evidence and Analysis

Scholars have highlighted how gig economy platforms transform employment relationships. Chan and Pun (2020) describe how Chinese food-delivery riders experience “platformization,” where digital tools coordinate all dimensions of their labor. Sun et al. (2021) provide survey evidence that riders in China average 9.8 hours of work per day, well above the statutory maximum of eight hours. This fact illustrates how algorithmic allocation normalizes overwork by rewarding long availability with better access to orders.

In Australia, UberEats reports present a different picture. According to Accenture and Uber (2021), riders in Sydney earned about A\$21.55 per hour during peak mealtimes, slightly above the statutory minimum wage of A\$20.33 set by the Fair Work Commission (2021). However, Stewart et al. (2018) caution that such figures exclude essential costs such as fuel, maintenance, and unpaid waiting time. Once these deductions are incorporated, riders’ effective hourly pay often falls below the legal minimum. In this case, algorithmic management does not lengthen hours as in China, but it still undermines wage protections by selectively framing data in ways favorable to the platform.

2.1 Algorithmic Control

Building on broader studies of digital management, Kellogg et al. (2020) describe algorithmic systems as “organizational experiments” that redistribute decision-making power away from workers. This framework helps explain how food-delivery platforms transform autonomy into control. A central feature of these platforms is algorithmic order allocation. Riders on UberEats must decide within seconds whether to accept a job; repeated refusals reduce their acceptance score, which lowers the chance of receiving high-value orders in the future (Rosenblat & Stark, 2016). The supposed freedom to reject work therefore becomes a mechanism of control. On Meituan, multi-order batching is routine: riders

are assigned several simultaneous deliveries to maximize efficiency. While this improves throughput, it compresses delivery windows and increases stress (Chan & Pun, 2020). In both systems, the algorithm disciplines behavior under the guise of flexibility, but the result is suppressed earnings and a cycle of low-value work that contributes to riders' persistent poverty.

2.2 Opportunity Cost

Economic theory emphasizes opportunity cost: the value of time spent in one activity is the foregone return from another. In food delivery, unpaid waiting at restaurants and idle time in hotspots constitute uncompensated labor. Sun et al. (2021) document that riders in China spend large portions of their day waiting for orders or for meals to be prepared, none of which is remunerated. In Australia, downtime similarly erodes hourly earnings. The Accenture report's gross pay statistics cannot capture these hidden costs, which directly reduce riders' welfare. Over time, this invisible labor pushes effective hourly pay below subsistence levels, reinforcing conditions of working poverty.

2.3 Information Asymmetry

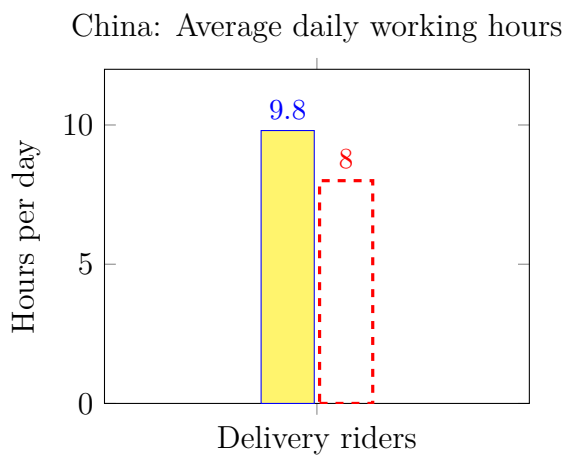
Information asymmetry further disadvantages workers. Platforms possess granular knowledge of demand, restaurant preparation times, and rider performance, but riders see only limited order details. As Rosenblat and Stark (2016) show in their study of Uber, this asymmetry allows platforms to shift risk: unattractive orders can be repeatedly offered until a rider accepts, while the rider cannot observe the broader allocation system. On Meituan, riders with high acceptance rates are more likely to be assigned remote or difficult orders, precisely because the platform assumes they will comply. This managerial opacity undermines the bargaining position of labor. This lack of transparency means riders cannot plan their work in ways that secure stable or adequate incomes, effectively institutionalizing poverty within the very architecture of algorithmic governance.

2.4 Externalization of Costs

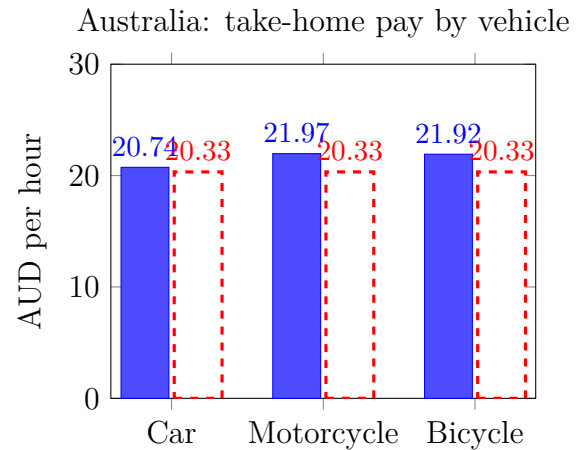
Platforms also externalize training and coordination costs. Referral bonuses incentivize riders to recruit new workers, while peer-to-peer guidance substitutes for formal training. Riders share tips on navigating the system, effectively performing unpaid managerial labor that sustains the platform (Chan & Pun, 2020). Similarly, riders bear all vehicle-related expenses. These strategies reduce platform liability while shifting the burden onto workers.

3 Visual Evidence

To anchor these arguments, stylized facts illustrate the contrast between China and Australia. In China, average rider hours (9.8 per day) exceed the statutory maximum of eight, underscoring how algorithmic design leads to normalized overwork. In Australia, Uber’s own data show gross hourly pay at A\$21.55, but independent analysis reveals that net returns often fall below the legal minimum. These facts motivate the claim that algorithmic management systematically undermines labor standards.



Source: Peer-reviewed surveys (Shanghai & Nanjing sample, 9.8h/day). Legal standard: 8 h/day (Labour Law, Article 36).



Source: Uber / Accenture report (Aug–Dec 2020, key mealtimes). Dashed red line = national minimum wage A\$20.33/h (FWC 2021).

The evidence points to a consistent pattern: algorithmic management reallocates risk from platforms to riders. By manipulating incentives, concealing allocation rules, and

excluding significant categories of labor time from compensation, platforms create a system where riders appear independent but functionally experience employer-like control. Wood et al. (2019) note that gig work globally exhibits this duality of autonomy and control: workers may choose when to log in, but once online they are governed by opaque algorithms.

Policy reforms are therefore urgent. In China, enforcement of statutory working-hour limits could mitigate excessive labor time. In Australia, regulators should require platforms to demonstrate that effective net pay, after deducting costs and accounting for downtime, meets minimum wage standards. Globally, mandates for algorithmic transparency would allow independent auditing and give riders greater agency in decision-making. Without such interventions, flexibility will remain a rhetorical device that conceals instability.

4 Conclusion

Returning to the broader literature on platform labor, the analysis here supports and extends prior findings (De Stefano (2016); Vallas and Schor (2020)) by demonstrating that algorithmic control not only reproduces traditional labor subordination but also deepens economic precarity across national contexts. This essay has argued that algorithmic management on food delivery platforms erodes labor standards through externalization of costs, information asymmetry, and disguised control. Stylized evidence from China and Australia demonstrates how these dynamics play out: in one case, overwork beyond legal limits; in the other, net wages falling below statutory minima. Far from empowering riders, platform algorithms consolidate managerial power while avoiding employer responsibilities. To reconcile efficiency with fairness, societies must adapt labor protections to the realities of algorithmic work. Without such reforms, food delivery platforms will continue to reproduce poverty by forcing riders to shoulder risks and costs while denying them the possibility of a dignified livelihood.

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